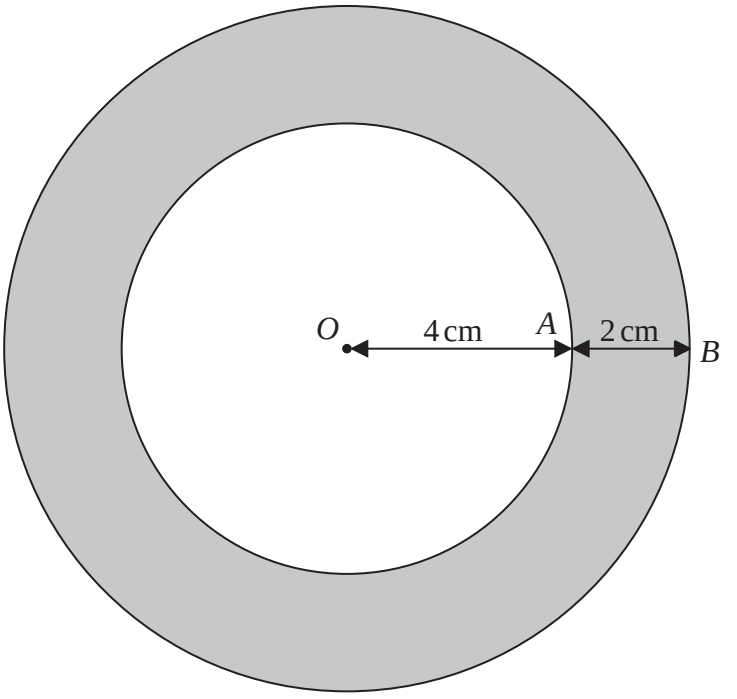
8 The diagram shows two circles.



Each circle has centre O.
OAB is a straight line.

Tim says,
"Exactly half of the diagram is shaded because $AB = \frac{1}{2} OA$"

Is Tim correct?
You must show all your working.

Outer radius = 4 + 2 = 6 cm

Outer area = $\pi \times 6^2 = 36\pi$ cm^2

1 mark
Use the two radii, 4 cm and 6 cm.

Inner area = $\pi \times 4^2 = 16\pi$ cm^2

Shaded area = $36\pi - 16\pi = 20\pi$ cm^2

1 mark
Subtract circle areas.

Half the whole area = 18π cm^2

$20\pi \neq 18\pi$, so Tim is NOT correct.

1 mark
20π is more than half of 36π.

(Total for Question 8 is 3 marks)

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